Regression tables

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.637814539
R Square	0.406807387
Adjusted R Square	0.399484021
Standard Error	0.196036444
Observations	100

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	2.134773225	2.134773225	55.54923909	8.88714E-11
Residual	99	3.112853281	0.038430287		
Total	100	5.247626506			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	0.480283003	0.384489681	1.249144064	0.21521022	-0.284730857	1.245296863	-0.284730857	1.245296863
x (SAT score)	0.001549208	0.00020786	7.453136192	8.88714E-11	0.001135632	0.001962784	0.001135632	0.001962784

Regression tables

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Multiple R	0.638
R Square	0.407
Adjusted R Square	0.399
Standard Error	0.196
Observations	100

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
SSE	1	2.135	2.135	55.549	0.000
SSR	99	3.113	0.038		
SST	100	5.248			

	Coefficients (b)	Standard Error	t Stat	P-value
Intercept	0.4803	0.3845	1.25	0.215
x (SAT score)	0.0015	0.0002	7.45	0.000

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The R-squared measures how much of the variability of the data is explained by your model

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The standard error is a correction of the standard deviation that takes into account the sample size

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$$\sqrt{\frac{SSE}{n-2}}$$

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The bigger the sample size, the lower the error, the better the model

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ANOVA = ANalysis Of VAriance

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$b_0 = 0.4803$

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$b_0 = 0.4803$ $b_1 = 0.0015$

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$b_0 = 0.4803$ $b_1 = 0.0015$ $\hat{y} = 0.4803 + 0.0015x$

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$H_0: \beta = 0$

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$H_0: \beta = 0$ is the coefficient equal to zero?

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If the coefficient of the intercept is 0, the line crosses the y-axis at the origin

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$H_0: \beta = 0$ is the coefficient equal to zero?

If β_1 is 0, than 0 times x will always be null for any x, so this variable will not be considered for the model